

Lecture 11: Trading with Linear Permanent Impact

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11.1 Linear Permanent Price Impact Mechanics

In previous lectures, price impact was assumed to be purely temporary—affecting only the immediate execution price while leaving the underlying mid-price process S_t unaffected once trading ceased. We now extend our microstructural framework to incorporate **permanent price impact**, where continuous execution persistently shifts the fundamental mid-price.

Let S_t denote the unaffected mid-price. Under linear permanent price impact with parameter $\gamma > 0$, continuous selling at rate $\nu_t = -\frac{d}{dt}Q_t \geq 0$ depresses the mid-price according to the controlled SDE

$$dS_t = -\gamma \nu_t dt + \sigma dW_t ,$$

where W_t is a standard Brownian motion and $\sigma > 0$ represents asset volatility.

Simultaneously, execution incurs local temporary price impact $f(\nu_t)$, yielding the effective per-share price $\hat{S}_t$,

$$\hat{S}_t = S_t - f(\nu_t) .$$

The cumulative cash process X_t generated by liquidating shares up to time t evolves as

$$dX_t = \hat{S}_t \nu_t dt = (S_t - f(\nu_t)) \nu_t dt , \quad X_0 = 0 .$$

11.2 Path-Independence & Duality to Terminal Penalty

A fundamental feature of linear permanent price impact is that its total expected impact cost is invariant to the trading trajectory $(Q_t, \nu_t)_{t \in [0, T]}$. This path-independence establishes a direct mathematical equivalence to a model with zero permanent impact and a shifted terminal inventory penalty.

Theorem (Path-Independence & Effective Terminal Penalty): *For any continuous execution strategy $\nu_t = -\frac{d}{dt}Q_t$ liquidating position $Q_0 = \mathfrak{N}$ down to Q_T over $[0, T]$, the total expected portfolio value (cash plus mark-to-market inventory) at time T satisfies*

$$\mathbb{E}[X_T + Q_T S_T] = Q_0 S_0 - \frac{\gamma}{2} Q_0^2 + \frac{\gamma}{2} \mathbb{E}[Q_T^2] - \mathbb{E}\left[\int_0^T f(\nu_t) \nu_t dt\right] . \quad (11.1)$$

Maximizing an objective with linear permanent impact γ and soft terminal penalty αQ_T^2 ,

$$J(\nu) = \mathbb{E} [X_T + Q_T S_T - \alpha Q_T^2]$$

is mathematically identical to maximizing an objective with **zero** permanent impact ($\gamma = 0$) and an effective terminal inventory penalty,

$$\alpha_{\text{eff}} = \alpha - \frac{\gamma}{2} .$$

Proof: Applying Itô's Lemma to the mark-to-market position value $Q_t S_t$:

$$d(Q_t S_t) = Q_t dS_t + S_t dQ_t = Q_t (-\gamma \nu_t dt + \sigma dW_t) - S_t \nu_t dt .$$

Rearranging for the gross revenue rate $S_t \nu_t dt$,

$$S_t \nu_t dt = -d(Q_t S_t) - \gamma Q_t \nu_t dt + \sigma Q_t dW_t .$$

Integrating from $t = 0$ to $t = T$,

$$\int_0^T S_t \nu_t dt = Q_0 S_0 - Q_T S_T - \gamma \int_0^T Q_t \nu_t dt + \int_0^T \sigma Q_t dW_t .$$

Substituting $\nu_t dt = -dQ_t$, the permanent impact integral evaluates directly as an exact differential,

$$\int_0^T Q_t \nu_t dt = - \int_0^T Q_t dQ_t = \frac{1}{2} Q_0^2 - \frac{1}{2} Q_T^2 .$$

Substituting this integral into total cash $X_T = \int_0^T S_t \nu_t dt - \int_0^T f(\nu_t) \nu_t dt$ gives

$$X_T + Q_T S_T = Q_0 S_0 - \frac{\gamma}{2} Q_0^2 + \frac{\gamma}{2} Q_T^2 - \int_0^T f(\nu_t) \nu_t dt + \int_0^T \sigma Q_t dW_t .$$

Taking expectations yields equation (11.1). Subtracting the soft terminal penalty $\alpha \mathbb{E}[Q_T^2]$ from both sides then yields

$$J(\nu) = Q_0 S_0 - \frac{\gamma}{2} Q_0^2 - \mathbb{E} \left[\int_0^T f(\nu_t) \nu_t dt \right] - \left(\alpha - \frac{\gamma}{2} \right) \mathbb{E} [Q_T^2] ,$$

which proves $\alpha_{\text{eff}} = \alpha - \frac{\gamma}{2}$. ■

Corollaries & Intuition:

1. **Sunk Cost under Hard Liquidation** ($Q_T = 0$): If complete liquidation is required, permanent impact contributes a fixed expected cost $\frac{\gamma}{2} \mathfrak{N}^2$, independent of whether liquidation is uniform, rapid, or irregular.
2. **Structural Rebate Mechanism:** Selling inventory early depresses the baseline mark-to-market price S_T at which remaining inventory Q_T is valued. This price depression reduces the mark-to-market penalty, acting as a structural rebate of $\frac{\gamma}{2} Q_T^2$ against the terminal penalty αQ_T^2 .
3. **Decoupling from Speed Profile:** Linear permanent impact does not alter the optimal trajectory shape ν_t^* , which remains dictated solely by temporary impact $f(\nu)$ and inventory risk aversion.

11.3 HJB Analysis & State Space Reduction

We verify this equivalence within the stochastic dynamic programming framework. Define the value function,

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t,s,q} \left[\int_t^T (S_u - f(\nu_u)) \nu_u du + Q_T S_T - \alpha Q_T^2 \right] .$$

The general Hamilton-Jacobi-Bellman (HJB) equation accounting for permanent price drift $-\gamma\nu$ is

$$\frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 H}{\partial s^2} + \max_{\nu} \left\{ (s - f(\nu)) \nu - \nu \frac{\partial H}{\partial q} - \gamma \nu \frac{\partial H}{\partial s} \right\} = 0 , \quad (11.2)$$

subject to terminal condition $H(T, s, q) = qs - \alpha q^2$.

Value Function Ansatz. Motivated by our path-independence result, we propose the additive decomposition

$$H(t, s, q) = qs - \frac{\gamma}{2} q^2 + h(t, q) . \quad (11.3)$$

Calculating partial derivatives,

$$\frac{\partial H}{\partial s} = q , \quad \frac{\partial^2 H}{\partial s^2} = 0 , \quad \frac{\partial H}{\partial q} = s - \gamma q + \frac{\partial h}{\partial q} , \quad \frac{\partial H}{\partial t} = \frac{\partial h}{\partial t} .$$

Elimination of Price Dependency. Substituting these derivatives into equation (11.2),

$$\frac{\partial h}{\partial t} + \max_{\nu} \left\{ s\nu - f(\nu)\nu - \nu \left(s - \gamma q + \frac{\partial h}{\partial q} \right) - \gamma \nu q \right\} = 0 .$$

Simplifying inside the maximum operator,

$$s\nu - f(\nu)\nu - s\nu + \gamma \nu q - \nu \frac{\partial h}{\partial q} - \gamma \nu q = -f(\nu)\nu - \nu \frac{\partial h}{\partial q} .$$

All state-dependent price terms s and permanent impact cross-terms γq cancel out identically! The PDE reduces to

$$\frac{\partial h}{\partial t} + \max_{\nu} \left\{ -f(\nu)\nu - \nu \frac{\partial h}{\partial q} \right\} = 0 . \quad (11.4)$$

Transformed Terminal Condition. At $t = T$, matching ansatz (11.3) to $H(T, s, q) = qs - \alpha q^2$ yields

$$qs - \frac{\gamma}{2} q^2 + h(T, q) = qs - \alpha q^2 \implies h(T, q) = - \left(\alpha - \frac{\gamma}{2} \right) q^2 .$$

This confirms that linear permanent impact is completely absorbed by reducing the terminal penalty coefficient to $\alpha_{\text{eff}} = \alpha - \frac{\gamma}{2}$.

11.4 Explicit Trajectories under Linear Temporary Impact

Under quadratic temporary impact $f(\nu) = k\nu$ ($k > 0$), the first-order condition in equation (11.4) gives optimal speed $\nu^* = -\frac{1}{2k} \frac{\partial h}{\partial q}$. Proposing $h(t, q) = -A(t)q^2$, $A(t)$ obeys the Riccati ODE

$$\frac{d}{dt}A(t) - \frac{1}{k}A(t)^2 = 0, \quad A(T) = \alpha - \frac{\gamma}{2}.$$

Solving this ODE gives

$$A(t) = \frac{\alpha - \frac{\gamma}{2}}{1 + \frac{\alpha - \frac{\gamma}{2}}{k}(T - t)}.$$

The corresponding optimal inventory trajectory Q_t^* and execution speed ν_t^* are

$$Q_t^* = \mathfrak{N} \left(\frac{1 + \frac{\alpha - \frac{\gamma}{2}}{k}(T - t)}{1 + \frac{\alpha - \frac{\gamma}{2}}{k}T} \right) \quad \text{and} \quad \nu_t^* = \frac{\mathfrak{N} \left(\frac{\alpha - \frac{\gamma}{2}}{k} \right)}{1 + \frac{\alpha - \frac{\gamma}{2}}{k}T}.$$

Regime Analysis:

- **Hard Liquidation** ($\alpha \rightarrow \infty$): $\frac{\alpha - \gamma/2}{k} \rightarrow \infty$, recovering standard TWAP execution: $Q_t^* = \mathfrak{N} \left(1 - \frac{t}{T} \right)$ and $\nu_t^* = \frac{\mathfrak{N}}{T}$.
- **Neutral Threshold** ($\alpha = \gamma/2$): $A(t) = 0 \implies \nu_t^* = 0$. The agent optimal strategy is to hold full inventory $Q_t^* = \mathfrak{N}$ perpetually.
- **Sub-Critical Penalty** ($\alpha < \gamma/2$): $A(T) < 0 \implies \nu_t^* < 0$. The permanent price displacement outweighs the terminal penalty, incentivizing the agent to buy additional shares rather than sell.

11.5 Remarks

1. **No-Arbitrage Constraint (Gatheral, 2010)**: Nonlinear permanent impact $dS_t = -\gamma\nu_t^\beta dt$ with $\beta \neq 1$ admits statistical price manipulation (round-trip trades yielding positive expected profit). Linear permanent impact ($\beta = 1$) is the unique specification compatible with market efficiency.
2. This lecture is based on Chapter 6.5 in [2], building upon foundational concepts introduced in [1] and [3].

References

- [1] R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–40, 2001.
- [2] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- [3] J. Gatheral. No-dynamic-arbitrage and market impact. *Quantitative Finance*, 10(7):749–759, 2010.